Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/18/2009
PI Name:	Putcha, Lakshmi Ph.D.		
Project Title:	Bioavailability & Performance Effects of Promethazine During Space Flight - DSO 490B		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	Deceased as of September 2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	96-OLMSA-01
Start Date:	10/01/2005	**End Date:**	09/30/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS 11, 13, 15, 16		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Harm, Deborah (NASA JSC)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Promethazine (PMZ) is used to treat space motion sickness (SMS) during Shuttle Missions. However, side effects associated with PMZ when used on Earth include dizziness, drowsiness, sedation, and impaired psychomotor performance, which could impact crew performance or mission operations. Early anecdotal reports from crewmembers indicate that these central nervous system side effects of PMZ are absent or greatly attenuated in microgravtity. Therefore, systematic evaluation of PMZ bioavailability, effects on performance, side effects and efficacy in the treatment of SMS are essential for determining optimal dosage and route of administration of PMZ in flight. OBJECTIVE: To establish dose-response relationship of PMZ before and during flight and estimate bioavailability of PMZ during flight. To compare and contrast the effect of flight-related variables (e.g. space motion sickness (SMS), mission duration, sleep medications, and crew activity schedules) and PMZ administration on sleep quality and

Task Description:	efficiency. RELEVANCE: This study will lead to a better understanding of how Promethazine is handled by the body in space. It will help to identify necessary changes for optimizing space motion sickness theraputics in space. BDC SUMMARY: At L-120 days, a 30-minute familiarization/training session will be held, with no data collection required. If promethazine was taken inflight, data collection identical to in-flight data collection will be done at R+30, with recording of sleep/wake times, timing of medication taken, saliva collection times, and sleepiness scoring. See also https://
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2006. [Ed. note: Task added to Task Book in March 2009]
Bibliography Type:	Description: (Last Updated: 06/23/2025)